2025_Hexaware_Actual_Test_12

Test Summary

- No. of Sections: 3
- No. of Questions: 92
- Total Duration: 130 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. What is the highest power of 5 that divides 90 x 80 x 70 x 60 x 50 x 40 x 30 x 20 x 10?

- 10
- 15
- 14
- 8

Q2. Find the least number which when divided by 5, 6, 7, 8 and 9 leaves reminders 3, 4, 5, 6 and 7 respectively.

- 2518
- 5036
- 1759
- 60

Q3. A can plough a farm at the same time as B and C together can do it. If A and B together do the same in 20 days and C in 100 days then B could do it alone in:

- 65 days
- 75 days
- 25 days
- 50 days

Q4. If 5 bakers with 9 assistants can bake 135 loaves of bread in 6 days and 3 bakers with 5 assistants can bake 75 loaves of bread in 4 days, find the time in which 7 bakers with 7 assistants will bake 525 loaves of bread.

- 3

5

7

10

Q5. A goods carriage of length 2 km headed to Srinagar from Punjab was running at a speed 30 km/hr. It crosses a tunnel which is 58 km long with that speed. Find the time taken by the goods carriage to cross the tunnel

4 hrs

2 hrs

3 hrs

1 hrs

Q6. Three persons walking around a circular track complete their respective single revolutions in 15(1/6) seconds, 16(1/4) seconds and 18(2/3) seconds respectively. They will be again together at the common starting point after 1 hour and:

10 seconds

20 seconds

30 seconds

40 seconds

Q7. In a test, a Alex could answer 70 % C++ questions, 40 % C questions and 60 % Java questions correctly. The test had a total of 75 questions, 10 from C++, 30 from C and 35 from Java. A minimum of 60 % in aggregate was required to be considered for interview. The Alex could not clear the test and was not shortlisted for interview. Find by how much marks did the Alex miss the interview call, given that each question was of 1 mark and there was no negative marking for incorrect answers.

4

5

6

7

Q8. In an examination, 80% of the students passed in English, 85% in Mathematics, and 75% in both English and Mathematics. If 40 students failed in both subjects, find the total number of students who appeared in the examination.

600

500

400

300

Q9. An article was sold at a profit of 12% . If the cost price would be 10% less and selling price would be Rs. 5.75 more, there would be a profit of 30 %. Then at what price it should be sold to make a profit of 20% = ?

Rs. 215

Rs. 138

Rs. 120

Rs. 115

Q10. A bookseller sells a book at a profit of 10%. Suppose he had bought it at 4% less and sold it for Rs. 6 more. He would have gained 75/4% . The cost price of the book is?

Rs. 160

Rs. 150

Rs. 140

Rs. 130

Q11. Find out two quantities whose difference is 30 and the ratio between them is 5/11.

20, 50

25, 55

30, 60

50, 110

Q12. Chandra and Naveen wish to buy a snack. Together they have Rs. 4 in total, consisting of quarters, dimes, and nickels. If they have 35 coins and the number of quarters is half the number of nickels, how many nickels and dimes do they have? (quarter = 0.25, dime = 0.1, nickel = 0.05)

5

20

10

25

Q13. In how many years an amount become double if the rate of interest given is 20%?

2 years

5 years

7 years

11 years

Q14. A person bought a motorbike under the following scheme: A down payment of Rs.15,000 and the rest amount at 8% per annum for 2 years. In this way, he paid Rs.28,920 in total. Find the actual price of the motorbike. (Assume simple interest).

Rs.26,000

Rs.27,000

Rs.27,200

Rs.26,500

Q15. In how many ways is it possible to make a selection by taking any number of 15 fruits, namely 3 oranges, 5 apples and 7 mangoes?

131

191

68

3720

Q16. How many words can be formed by using all the letters of the word 'ALLAHABAD'?

1890

7560

2520

3780

Q17. If two dice are thrown simultaneously what is the probability of getting a sum less than 10

1/6

4/5

5/6

10/12

Q18. Three dice are thrown together. Find the probability of getting a total of at least 6.

103 / 216

103 / 208

103 / 108

36 / 108

Q19. The first term of an AP is 10 and the last term is 28. If the sum of all terms is 190, what is the common difference?

5

3

1

2

Q20. If the first two terms of a harmonic progression are 1/16 and 1/13, find the maximum partial sum.

1.63

1.13

1.89

1.22

Q21. Ajay walks 24 km towards the East turns to the right-hand side and takes a drive of another 10 km. He then turns to his right (drives towards the West) another 10 km. He then turns to his left & walks another 8 km. After that, he turns to his right & travels 14 km. How far is he from his initial point & in which direction?

20 km East

18 km south

16 km West

10 km South

Q22. Sailaja started from her house and walked 11 km towards west followed by 20 km to her right. She then took a right turn to walk 6 km. Finally she walked 8 km to her right. What is the distance between Sailaja's house and her destination?

17 km

12 km

15 km

13 km

Q23. The question below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both statements and give an answer.

The number of cards I have lies between 110 and 120. What is the exact number?
(A) When I count them 3 at a time, I have 2 left over.
(B) When I count them 5 at a time, I. have 1 left over.

if statement (A) alone is sufficient to solve the question, but statement (B) alone is not.

if statement (B) alone is sufficient to solve the question, but statement (A) alone is not.

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

if both the statements taken together are not sufficient and more information is required to solve the question.

Q24. Two statements numbered I and II are given below. You have to decide whether the data provided in the statements are sufficient to answer the question.

What is the code for 'juicy apple'?

Statements:
I. The code for 'apple juice is cedar' is 'ka la do ni', the code for 'cedar vinegar is good' is 'do re la mi' and the code for 'buy some juicy apples' is 'vu hi wo pe'.
II. The code for 'good orange and apple' is 'ni xe co mi', the code for 'apple juice is fresh' is 'ka la ni po' and the code for 'fresh and juicy orange' is 'xe co pe po'.

If the data in statement I alone are sufficient to answer the question while data in statement II alone is not sufficient.

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

If the data in both statements I and II together are not sufficient to answer the question

If the data either in statement I alone or in statement II alone is sufficient to answer the question.

If the data in both statements I and II together are necessary to answer the question

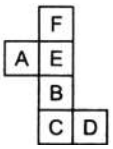
Q25. **Directions:**
What will come in place of question mark (?) in the following series?

582, 751, 555, 724, 528,?.

Q26. **Directions:**
What will come in place of question mark (?) in the following series?

13, 15, 21, 37, 77, 173, ?.

Q27. In a given figure of open dice. Which dice we can make?











Q28. Observe the dots on the dice (one to six dots) in the following figures. How many dots are contained on the face opposite to the containing four dots?





2

3

5

6

Q29. **Find the missing term in the series:**

QAR, RAS, SAT, TAU, _____.

UAV

UAT

TAS

TAT

Q30. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

A. But the situation has changed in recent times and we are seeing judges getting involved in corruption.
B. In this changed scenario, a special committee must be constituted to look into the affairs of the judiciary.
C. Indians have great faith in the Indian judicial system.

D. Due to this people are losing faith in the judiciary.
E. They consider the judiciary as supreme.

- BCEDA
- ABCED
- CEADB
- ECADB

Q31. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

A. Reports of overcrowded classrooms, outdated facilities, drug use, and violent behavior can be found in newspapers across the country.
B. What then is the future of public education?
C. In addition, many parents have grown increasingly concerned by the curriculum offered to their children, raising questions about textbooks that teach evolution and libraries that contain books challenging traditional values.
D. Can we recommit ourselves as a nation to the belief that every child is entitled to, a free public education that ensures literacy and civic responsibility?
E. Although the quality of public education varies from one school district to another and often among schools in the same district, many Americans are beginning to believe good schools are rare.
F. Or do we decide that public schools as we have known them are ill-suited to the needs of a new century?

- ABDFEC
- EACBDF
- DBAECF
- AEDFCB

Q32. Choose the word opposite in meaning to the given word

Gull

- Dupe
- Swindle
- Hoax
- Nourish

Q33. **Select the option that is most nearly OPPOSITE in meaning to the given word:**

BLASÉ (OPPOSITE)

- fiery
- clever
- intriguing

slim

ardent

Q34. **Select the word or phrase which best expresses the meaning of the given word:**

BLITHE

Disturb

Carefree

Distress

Emotive

Q35. **Fill the blank with appropriate conditionals.**

We _____ watched TV tonight if Peter hadn't bought the theatre tickets.

could have

should have

would have

might have

Q36. I did my homework while I _____television.

had watched

was watching

watching

was watch

Q37. **Replace the underlined part of the sentence with a grammatically correct option.**

Does he thought about the consequences if he refuses the offer of someone so powerful?

Does he thought about

Does he know

He will have to face

Had he thought about

Q38. **In each of the following sentences, a part of the sentence or the whole sentence is underlined. Beneath each sentence, five different ways of phrasing the underlined part are indicated. Choose the best alternative from among the five.**

In one of the most surprising victories in World War I, the newly-formed Soviet communist state was routed by the Polish General, Jozef Pilsudski, demanding that it should push back its borders east of Vilnius.

- demanding that it should
- demanding it to
- and their demand to
- who demanded that it
- who demanded them to

Q39. **Fill in the blank with a suitable article from the options given.**

She returned after ..... hour.

- a
- an
- the
- no article required

Q40. **Fill up the blank with suitable prepositions and conjunctions from the alternatives given under each sentence.**

No sooner had I reached the office _____ I got the news of your arrival.

- and
- when
- than
- since

Common Content:

Eight friends M, N, O, P, Q, R, S and T are sitting in a straight line facing north. All of them like different games, viz Cricket, Badminton, Hockey, Golf, Football, Baseball, Chess and Tennis, but not necessarily in the same order. The one who likes Tennis sits third to the right of S. O is on the immediate left of S. The person who likes Hockey sits second to the left of O. N sits second to the right of T, who likes neither Tennis nor Hockey. Only two persons sit between O and the one who likes Football. M and R immediate neighbours and both of them do not like Tennis. The person who likes Cricket sits second to the right of M. There are two persons between P and the one who likes Badminton. P does not like Tennis. The one who likes Baseball is an immediate neighbour of the one who likes Tennis. The one who likes Football sits second to the left of M. One of the immediate neighbours of S likes Chess. P sits on an extreme end of the line.

Q41. Who among the following sits on the immediate left of R?

- O
- M

P

None of these

Q42. What is the position of N with respect to the one who likes Chess?

Second to the left

Second to the right

Third to the right

Fourth to the left

Q43. How many persons are there between M and T?

Three

Two

None

Four

Q44. O likes which of the following game?

Badminton

Cricket

Baseball

None of these

Q45. Who among the following likes Golf?

M

O

R

P

Common Content:

Study the information given below and answer the questions following it:

'Move Fast Or Left Behind' is coded as '7 1 9 0 3'
'Move Left Behind The Journey' is coded as '4 9 1 5 3'
'Your Journey Ended Fast' is coded as '5 2 7 6'
'The Life Ended Or Behind' is coded as '0 8 4 9 2'

Q46. What is the code for "left"?

5

6

2

1

Q47. Which word has been coded as “3”?

Move

Behind

The

Journey

Q48. How will ‘your life journey’ be coded as?

5 6 8

1 2 3

9 0 6

0 2 5

Q49. What is the code for ‘fast’?

5

0

6

7

Common Content:

Read the following information carefully and answer the questions given below
A # B means ‘A is daughter of B’
A $ B means ‘A is husband of B’
A * B means ‘A is brother of B’
A @ B means ‘A is sister of B’

Q50. If M * P # Q @ R, which of the following is definitely true?

P is the cousin of R

M is the nephew of P

Q is the uncle of M

None is true

Q51. If P # Q $ R @ T, which of the following is true?

T is the sister of R

R is an aunty of P

T is the brother-in-law of Q

P is the daughter of R

Common Content:

Read the passage and answer the following questions:

Teachers should be aware of the importance of their role in helping gifted students in social and emotional adjustment, and in promoting their mental health. Many signs of maladjustment appear at quite an early age and teachers have the opportunity to observe students under conditions which disclose such behaviour. If need be, expert help should be obtained in treating such maladjustment. When parental relations are strained, the child is likely to feel insecure. Such a child can be helped by giving him some individual attention and making him feel that he can win the teacher's respect and affection for his superior performance.

When discipline in the home is weak and inconsistent, the teacher can do much by providing the child opportunities for good behaviour. Bright students are as quick to respect firm and consistent discipline as they are to challenge one that is lax. Bright students who misbehave at home may welcome stricter discipline at school and they may in fact be grateful for the support of a firm hand in their efforts to control themselves.

The gifted child may need help in human relations. He cannot but feel different from his average classmates. If he can find no companions who can share his interests and concerns, he may begin to stay apart more and more. One result may be a tendency to solitary pursuits such as reading, collecting things or scientific experimentation. He misses group games and rough and tumble type of play. He falls behind in physical skills and when, by chance, he finds himself in games or group activities he feels awkward and inadequate. The teacher could help by giving him the opportunity of leadership in the activities in which he is superior, by asking him to help the weaker students in studies and so on.

Q52. A child is likely to feel insecure when

strict discipline is imposed.

he does not have friends.

he has no money.

relationship between parents is strained.

Q53. A child who has no companion enjoys

physical games.

fine arts.

reading or collecting things.

only intellectual activities.

Q54. The passage focuses on

below average students.

the role of discipline in schools.

the mental health of the gifted students.

parent-child relationship.

Q55. The gifted child needs help in human relations because he is

predisposed to solitary pursuits.

different from his classmates and cannot share his interests with them.

emotionally imbalanced.

inadequate in physical activities.

Q56. The teacher's role in helping the gifted, child is important because he/she

has the opportunity to notice maladjustments in a child.

can impose discipline on students.

can cure all problems in children.

is always with children.

Common Content:

In the following passage, there are blanks, each of which has been numbered. Choose words that complete the passage appropriately.

Agriculture has always been celebrated as the primary sector in India. Thanks to the Green Revolution, India is now ___(1)____ in food production. Indian agriculture has been ____(2)___ advancement as well. Does that mean everything is looking bright for Indian agriculture? A superficial analysis of the above points would tempt one to say yes, but the ___(3)_____. The reality is that Indian farmers have to face extreme poverty and financial crisis, which is _____(4)___ suicides. What are the grave adversities that drive the farmers to commit suicide? At a time when the Indian economy is supposed __(5)____ to take on the world?

Q57. Fill in the blank numbered(1)

perfect about

rely to food

self-sufficient

dependent to food

Q58. Fill in the blank numbered(2)

longing to greenery

making technological

creating marginal

producing grains

Q59. Fill in the blank numbered(3)

reality suggests the same

demand is same

reality is bright

truth is far from it

Q60. Fill in the blank numbered(5)

to alleviate up

to imagined for elevation

to be gearing up

to gradually steeping up at

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. Which of the following is a private IPv4 address range?

172.16.0.0 to 172.31.255.255

192.168.0.0 to 192.168.255.255

10.0.0.0 to 10.255.255.255

All of the mentioned options

Q2. How many bits are there in an IPv6 address?

32 bits

64 bits

128 bits

256 bits

Q3. What is the primary motivation for the development of IPv6?

- Faster data transmission
- Increased security
- Larger address space
- Improved routing algorithms

Q4. What is the purpose of the Network Layer in the OSI model?

- Physical Transmission
- End-to-End Communication
- Data Link Establishment
- Error Detection

Q5. What does the SYN flag indicate in a TCP segment?

- Synchronize sequence numbers
- Finish the connection
- Acknowledge received data
- Congestion Avoidance

Q6. Which of the following is true about UDP?

- It guarantees the delivery of data
- It is connection-oriented
- It is faster than TCP
- It uses a three-way handshake for connection establishment

Q7. Which of the following is a feature of UDP but not of TCP?

- Connection-oriented
- Reliable Data Transfer
- Flow Control
- Low Overhead

Q8. Which of the following is a feature of TCP but not of UDP?

- Connectionless
- Reliable Data Delivery
- Low Overhead
- Broadcasting

Q9. Which of the following is a connectionless protocol at the Transport Layer?

- TCP
- IP
- UDP
- ARP

Q10. What is the primary purpose of the Transport Layer in the OSI model?

- Network Access
- End-to-End Communication
- Routing
- Physical Transmission

Q11. A DNS client is called?

- DNS updater
- DNS resolver
- DNS handler
- None of the mentioned options

Q12. What is the purpose of ARP (Address Resolution Protocol)?

- IP Address Assignment
- Name Resolution
- MAC Address Resolution

Dynamic Routing

Q13. In which layer of the OSI model does FTP operate?

Data Link Layer

Network Layer

Transport Layer

Application Layer

Q14. What is the default port for DNS?

53

21

25

80

Q15. What will be the output of the following pseudocode when a=12 and b=6?

```
Integer funn(Integer a, Integer b)
  if(a+10>b || a>20)
    return funn(a-10, a+10)
  End if
  return a + b + 10
```

34

32

36

None of the mentioned options

Q16. What will be the output of the following pseudocode for a=5 and b=2?

```
Integer funn(Integer a, Integer b)
  if (b<a)
    a=1
    b=b+a
    return funn(a,b+a)
  End if
return a+b+1
```

9

3

6

14

Q17. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=4, q=2, r=7
3  q= (7+5)+r
4  r=(p+5)+r
5  if((q+5)<(5-q))
6    p=(r+r)+q
7    r=p+p
8  End if
9  Print p+q+r
```

41

49

36

39

Q18. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=2,b=4,c=10
3  b=(7^12)&a
4  b=9&a
5  if ((c+b)<(10-c))
6      b=2^c
7      b=(c^a)+b
8      b=(2+12)^b
9  End if
10 Print a+b+c
```

12

27

13

7

Q19. What will be the output of the following pseudocode for a=2 and b=8?

```
1  Integer funn(Integer a, Integer b)
2     if (a+10>b || a>20)
3        return funn(a+10,a-10)
4     End if
5     return a+b+10
```

24

11

14

None of the mentioned options

Q20. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set p=0, q=4, r=8
3  for(each r from 3 to 7)
4    if((q-r)<(r+q))
5      jump out of the loop
6    End if
7    p=(6+3)+r
8    p=(q+r)+r
9  End for
10 Print p+q
```

2

8

4

10

Q21. What will be the output of the following pseudocode for a=1 and b=4?

```
1  Integer funn(Integer a, Integer b)
2    if (2>a)
3      return funn(a+1,a+2) + funn(a+3,a+4)
4    End if
5  return a+1
```

8

5

13

20

Q22. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=1, q=9, r=10
3  p=(12 & 8) + p
4  q= (p^6) & p
5  r = 2 +q
6  Print p+q+r
```

37

29

25

46

Q23. What will be the output of the following pseudocode for a=4?

```
function(input a){
   if (a>=1)
      return a*function(a-1);
   else
      return 1;
}
```

18

24

27

6

Q24. What will be the output of the following pseudocode?

```
Integer pp, qq, rr
Set pp = 2, qq= 5, rr = 6
qq = (6&4) + rr
qq = (10&6) + pp
pp = 9 + rr
Print pp + qq + rr
```

3

41

25

30

Q25. What will be the output of the following pseudocode for a=2, b=3, and c=0?

```
Integer funn(Integer a, Integer b,Integer c)
   for (each c from 2 to 3)
      b=(1+1)^a
   End for
   a=(4+6)^a
   return a+b
End function funn()
```

203

109

104

101

Q26. Which of the statements given in the options is correct regarding the following pseudocode?

```
if(x > y):
   if(z > y):
      if(z == x):
         print ("two");
      else:
         print ("one");
   else:
      print ("three");
else:
   print ("four");
```

It will print one if x=15, y =110, z=92

It will print one if x=5, y =10, z=9

It will print three If x=9, y=8, z=10

It will print three If x=9, y=8, z=6

Q27. What will be the output of the following pseudocode?

```
Integer a, b, c, d
Set b=343
for (each a from 1 to 4)
   c=b MOD 10
   d=b/10
   a=a+c+d
end for
Print a
```

45

41

39

None of the mentioned options

Q28. What will be the output of the following pseudocode?

```
Integer p, q, r, s, t
Set q = 81, r = 10
while (q>r)
   p = q + r
   s = p / 10
   t = s * s
   q = q / 3
end while
if (t > s)
   Print s
else
   Print t
end if
```

9

12

3

6

Q29. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  Set a=8, b=2, c=4
3  if((a^c)<b))
4   if((a+c)<(c-a))
5     a=9+c
6     if((c+8)<(8-c))
7       c=(c+c)+b
8       c=c+a
9     End if
10    End if
11  End if
12  b=(4^3)+c
13  Print a+b+c
```

30

33

23

20

Q30. What will be the output of the following pseudocode for n = 243?

```
1  Integer abc (int n, int b)
2     Integer a, b
3     if (n > 0)
4        a = n mod 10
5        b = b + a
6        return abc(n/10, b)
7     else
8       return b
9  End abc ()
```

2

9

3

342

Section 3 - Coding

Section Summary

- No. of Questions: 2
- Duration: 40 min

Additional Instructions:

None

Q1. **Problem Statement**

You are given an integer array **coins** representing coins of different denominations and an integer amount representing a total amount of money. Return the fewest number of coins that you need to make up that amount. If that amount of money cannot be made up by any combination of the coins, return -1. You may assume that you have an infinite number of each kind of coin.

Example 1

Input:
1 2 5
11
Output:
3
Explanation:
11 = 5 + 5 + 1

Example 2

Input:
2
3
Output:
-1

Input Format

The first line of input consists of denominations of coins separated by spaces.
The second line consists of the amount for which you want to make the change.

Output Format

The output displays the number of coins needed to make up the given amount.

Refer to the sample output for formatting specifications.

Constraints

$1 \le coins.length \le 12$
$0 \le amount \le 10^4$

Sample Input

```
1 2 5
11
```

Sample Output

```
3
```

Sample Input

```
1
0
```

Sample Output

```
0
```

Sample Input

```
2
3
```

Sample Output

```
-1
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

Given two strings, **word1, and word2**, return the minimum number of operations required to convert word1 to word2.

You have the following three operations permitted on a word:

1. Insert a character
2. Delete a character
3. Replace a character

Example 1
Input:
horse
ros

Output:
3

Explanation:
horse -> rorse (replace 'h' with 'r')
rorse -> rose (remove 'r')
rose -> ros (remove 'e')

Example 2
Input:
intention
execution

Output:
5

Explanation:
intention -> inention (remove 't')
inention -> enention (replace 'i' with 'e')
enention -> exention (replace 'n' with 'x')
exention -> exection (replace 'n' with 'c')
exection -> execution (insert 'u')

Input Format

The input consists of two strings, word1, and word2, which represent the two words for which the minimum edit distance needs to be calculated.

Output Format

The output displays the minimum edit distance between the two words.

Refer to the sample output for the formatting specifications.

Constraints

In the given scenario, the test cases will fall under the following constraints:
0 ≤ word1.length, word2.length ≤ 100
word1 and word2 consist of lowercase English letters.

Sample Input

```
horse
ros
```

Sample Output

```
3
```

Sample Input

```
intention
execution
```

Sample Output

```
5
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Aptitude

Q1

10

Solution

Highest power of 5 is

90= 5*18

80= 5* 16

70= 5*14

60= 5*12

50=5^2*2

40=5*8

30=5*6

20=5*4

10=5*2

There are 10 fives

Q2

2518

Solution

LCM of 5, 6, 7, 8, 9 = 2520.

Now the difference in the divisor and remainder for all 5 numbers are constant = 2.

Hence, Answer = 2520 –2=2518.

Q3

50 days

Solution

$Let's\ assume\ that\ A,\ B,\ and\ C\ can\ plough\ the\ farm\ in\ a,\ b,\ and\ c\ days\ respectively\ if\ they\ work\ alone.$

$From\ the\ first\ statement,\ we\ can\ write: \frac{1}{a} = \frac{1}{b} + \frac{1}{c} \ldots (1)$

$From\ the\ second\ statement,\ we\ can\ write: \frac{1}{a} + \frac{1}{b} = \frac{1}{20} \ldots (2)$

$From\ the\ third\ statement,\ we\ can\ write: \frac{1}{c} = \frac{1}{100} \ldots (3)$

$Substituting\ this\ value\ in\ equation\ (1),\ we\ get: \frac{1}{a} = \frac{1}{b} + \frac{1}{100}$

$Substituting\ this\ value\ in\ equation\ (2),\ we\ get:$

$To\ solve\ the\ equation\ \frac{1}{b} + \frac{1}{100} + \frac{1}{b} = \frac{1}{20}\ for\ the\ value\ of\ b,$
$we\ can\ first\ simplify\ the\ left - hand\ side\ of\ the\ equation\ by\ finding\ a\ common\ denominator:$
$\frac{1}{b} + \frac{1}{100} + \frac{1}{b} = \frac{1}{b} + \frac{1}{b} + \frac{1}{100} = \frac{2}{b} + \frac{1}{100}$
$Now,\ we\ can\ substitute\ this\ expression\ back\ into\ the\ original\ equation: \frac{2}{b} + \frac{1}{100} = \frac{1}{20}$
$Multiplying\ both\ sides\ of\ the\ equation\ by\ the\ least\ common\ multiple\ of\ the\ denominators,\ which\ is\ 100b,$
$we\ get: 200 + b = 5b$
$Simplifying\ this\ expression,\ we\ get: 4b = 200 \Rightarrow b = 50$
$Therefore,\ the\ value\ of\ b\ is\ 50.$

Q4

5

Solution

5 bakers with 9 assistants can bake 135 loaves of bread in 6 days.

Meals baked in 1 day by 5 bakers and 9 assistants:

Daily loaves= 135 loaves/6 days = 22.5 loaves per day

3 bakers with 5 assistants can bake 75 loaves of bread in 4 days.

Meals baked in 1 day by 3 bakers and 5 assistants:

Daily loaves= 75 loaves/4 days =18.75 loaves per day

Let B be the number of loaves one baker can bake in 1 day, and

A be the number of loaves one assistant can bake in 1 day.

5B+9A=22.5

3B+5A=18.75

Multiply (i) by 3 and (ii) by 5 to align the coefficients for elimination:

15B+27A=67.5(iii)

15B+25A=93.75(iv)

Subtract (iii) from (iv):

(15B+25A)−(15B+27A)=93.75−67.5

−2A=26.25

A=− 26.25/2=−13.125 loaves per day

Substitute A back into equation (i) to find B:

5B+9(−13.125)=22.5

B=28.125 loaves per day

Use B and A to find the production rate for 7 bakers and 7 assistants:

7B+7A

=105 loaves per day

Time= 525 loaves/105 loaves per day

Time=5 days

Q5

2 hrs

Solution

train covered the distance of tunnel ,as well as its length.

So, the total distance travelled by the train is= 58 km + 2 km=60 km

now time taken =60 km / (30 km/hr)= 2 hours

Q6

40 seconds

Solution

Time taken by the persons to complete single revolutions = 15(1/6) s, 16(1/4) s, 18(2/3)s

= 91/6 , 65/4 , 56/3

Time required they will be again together at the common starting point

=LCM[91/6,65/4,56/3] [LCM of fraction =(LCM of Numerator/HCF of Denominator)]

=3640 sec.

=1 hour 40 sec

Q7

5

Solution

We are given that the Alex could answer 70 % C++ questions, 40 % C questions and 60 % Java questions correctly and there were total 75 questions : 10 from C++, 30 from C and 35 from Java.

=> C++ questions answered correctly = 70 % of 10 = 7

=> C questions answered correctly = 40 % of 30 = 12

=> Java questions answered correctly = 60 % of 35 = 21

=> Total questions answered correctly = 7 + 12 + 21 = 40

=> Marks secured = 40 x 1 = 40

Now, marks required = 60 % of 75 = 45

=> Shortfall in marks = 45 – 40 = 5

Therefore, the geek missed the interview call by 5 marks.

Q8

400

Solution

Let the total number of students be 100 n.

=> Students passed in English = 80 % of 100 n = 80 n

=> Students passed in Mathematics = 85 % of 100 n = 85 n

=> Students passed in both English and Mathematics = 75 % of 100 n = 75 n

=> Total number of students passed in at least one subject = 80 n + 85 n – 75 n = 90 n

=> Number of students who failed in both the subjects = 100 n – 90 n = 10 n = 40 (given)

=> n = 4

Therefore, the total number of students who appeared in the examination = 100 n = 400

Q9

Rs. 138

Solution

Let the CP1 of article = 100x

Initial SP1 = 100x + 12% of 100x = 100x + 12x = 112x

Now, If He brought article at 10% discount. Therefore CP2 = 90x

Now SP2 = 90x + 30% of 90x = 90x + 27x =117x

According to question

SP2 - SP1 = 5.75

⇒ 117x - 112x = 5.75

⇒ 5x = 5.75

⇒ x = 5.75/5

⇒ x = 1.15

Initial Cost of article = 100 × 1.15 = Rs. 115

To sold article at 20% profit

= 115 + 20% of 115

= 115 + 23

= Rs. 138

Q10

Rs. 150

Solution

Let the CP of the book = x

Gain = 10%

SP =110x/100

If he had bought it at 4% less and sold it for Rs 6 more,

CP=96x/100

SP=(110x/100)+6

Gain=(110x/100+6)−96x/100=(14x/100)+6

Now, the gain =75/4%

=>14x/100+6=96x/100×75/400

⇒4x=600∴

x=150

Q11

25, 55

Solution

The difference of quantities which are in the ratio 5 : 11 is 6.

To make the difference 30, we should multiply them by 5.

Therefore 5 : 11 = 5 × 5 : 11 × 5 = 25 : 55.

The quantities are 25 and 55.

Q12

25

Solution

No. of quarters = x

No. of nickels = 2x

No. of dimes = 35 - 3x

Values of them are :

0.25x + (0.05)2x + (0.1)(35-3x) = 4

0.05x = 0.5

x=10

Nickels + Dimes = 2(10) + 35-3(10)

= 25

Q13

5 years

Solution

Let the principal amount be P, Simple interest amount is also P (This is because the money gets doubled if half the doubled amount will be the principal amount then the remaining amount will be its interest). Hence, the interest and the principal amount is equal here.

Rate at which the amount increases for every year is 20 for the money to be doubled at 20%.

Number of years be taken as “T”

Therefore

P = PTx20/100

P=PxT/5

T = 5 years

Q14 Rs.27,000

Solution

15,000 + P(1 + (NR/100)) = 28920

15000 + P(1 + (16/100)) = 28920

p(1.16) = 13920

P = 12,000

motorbike cost 15000 + 12000 = Rs. 27000.

Q15 191

Solution

Out of 15 fruits, 7 are alike of one kind, 5 are alike of a second kind and 3 are alike of a third kind.

Hence, the required number of ways. = ((7 + 1) (5 + 1) (3 + 1) – 1] = 191.

Alternatively:

0 or more oranges can be selected from 7 identical mangoes in (7+1) = 8 ways

0 or more apples can be selected from 5 identical apples in (5+1) = 6 ways

0 or more mangoes can be selected from 3 identical oranges in (3+1) = 4 ways

Total number of ways in which all three types of fruits can be selected

(the number of any type of fruit may also be 0)= 8×6×4= 192

But in these 192 selections, there is one selection where all fruits are 0. Hence we need to reduce this selection.

Hence, required number = 192– 1 = 191

Q16 7560

Solution

The word 'ALLAHABAD' contains 9 letters, namely 4A, 2L, 1H, 1B and 1D

Required number of words = 9!/(4! * 2!) = 7560

Q17 5/6

Solution

Total outcomes = 6^2 = 36

Sample space of getting a sum greater or equal to 10

10 = (4,6) (6,4) (5,5)

11 = (5,6) (6,5)

12 = (6,6)

Required probability = 1-6/36 = 5/6

Q18

103 / 108

Solution

Since one die can be thrown in six ways to obtain any one of the six numbers marked on its six faces

⇒ Total number of elementary events = 6 x 6 x 6 = 216

Let A be the event of getting a total of at least 6. Then $\bar{A}$ denotes the event of getting a total of less than 6 i.e. 3, 4, 5.

⇒ $\bar{A}$ = { (1,1,1), (1,1,2), (1,2,1), (2,1,1), (1,1,3),(1,3,1), (3,1,1), (1,2,2), (2,1,2), (2,2,1) }

So, favorable number of cases = 10

⇒ P($\bar{A}$) = 10 / 216

⇒ 1 – P (A) = 10 / 216

⇒ P(A) = 1 – 10 / 216

= 103 / 108

Q19

2

Solution

The sum of all terms is (n/2) (a + l) = 190. This gives us n = 10.

The 10th term is (a + 9d) = 28. Substituting for a, we get d = 2

Q20

1.63

Solution

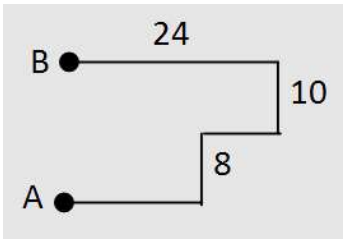
The terms of the HP are 1/16 , 1/13 , 1/10 , 1/7 , 1/4 , 1/1 , 1/-2 , 1/-5

So the maximum partial sum is 1/16 + 1/13 + 1/10 + 1/7 + 1/4 + 1/1 ≈ 1.63

Q21

18 km south

Solution

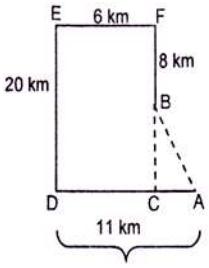


AB =10 + 8 = 18 South

Q22

13 km

Solution



Let A be the starting point and B the destination

The distance required is AB = $\sqrt{(BC^2 + AC^2)}$

BC = CF – FB (or) DE – FB = 20 – 8 = 12 km

AC = AD – CD (or) AD – EF = 11 – 6 = 5 km

∴ AB= $\sqrt{(12^2 + 5^2)} = \sqrt{(169)}$ = 13 km

Q23

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

Solution

From statement (A): The, number of cards could be 113,116 or 119.

From statement (B): The number of cards could be 111 or 116.

Combining statements (A) and (B): The number of cards has to be 116.

Q24

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

Solution

From I alone:

apple juice **is cedar** == ka **la do** ni

cedar vinegar **is** good == **do** re **la** mi

buy some juicy apples == vu hi wo pe

So, we get:

ls/cedar == la/do (not necessarily in the same order)

Apple/juice ==ka/ni (not necessarily in the same order)

Vinegar/good == re/mi (not necessarily in the same order)

So, the code for 'juicy apple' cannot be determined.

So, statement I alone is not sufficient.

From II alone:

good *orange and* **apple** == **ni** *xe co* mi

apple juice is **fresh** == ka la **ni po**

fresh *and* juicy *orange* == xe co pe **po**

so, we get

Apple == ni

And/ orange == xe/co (not necessarily in the same order)

Fresh == po

so, juicy == pe

So, the code for 'juicy apple' is 'ni pe'.

So, statement II alone is sufficient.

Q25 697

Solution

Series Pattern Given Series

582 582

582 + 169 = 751 751

751 – 196 = 555 555

555 + 169 = 724 724

724 – 196 = 528 528

528 + 169 = 697 697

Q26 397

Solution

Series Pattern Given Series

13 13

13 + 2 × 1 = 15 15

13 + 22 × 2 = 21 21

13 + 23 × 3 = 37 37

13 + 24 × 4 = 77 77

13 + 25 × 5 = 173 173

13 + 26 × 6 = 397 397

Q27



Solution

When we fold the question figure, We get 'B' as base of the cube and 'F' as Top, Thus they face each other.

'C' will face 'E' and 'A' will face 'D'.

Thus ,(B,F) ,(C,E) and (A,D) cannot be the adjacent sides of the cube.

So,

Figure 2 is possible.

Q28 2

Solution

We shall assume the dice in fig. (ii) to be rotated so that the 5 dots appear at the same position as in fig. (i) i.e. on RHS face (i.e. on face II as per activity 1) and 1 dot appears at the same position as in fig; (i) i.e. on Front face (i.e. on face I). Then, from the, two figures, 2 dots appear on the top face (i.e. on face V) and 4 dots appear on the Bottom face (i.e. on face VI).

Since, these two faces are opposite to each other, therefore, two dots are contained on the face opposite to that containing four dots.

Q29 UAV

Solution

In this series, the third letter is repeated as the first letter of the next segment. The middle letter, A, remains static. The third letters are in alphabetical order, beginning with R.

Q30 CEADB

Solution

The words - 'but', 'in this changed scenario', 'due to this', and 'they' in 'A', 'B', 'D' and 'E' respectively do no claim for the first position.

Indians and their faith in the judicial system is 'what is discussed in all other sentences and hence 'C' is the topic sentence and the opening sentence of the paragraph.

The CEADB is the correct sequence.

Q31

EACBDF

Solution

'E' begins the paragraph by introducing the topic - the growing realization that the quality of the public education is not good enough. 'A' continues as it talks about school-related news and how it is negative in nature. 'C' follows 'A' as it talks about another concern, when it comes to public education. 'B' follows as 'What then', is in the light of the concerns expressed in 'A' and 'C'. 'D' and 'F' form a pair as they both talk about the future of public. 'F' comes after 'D' as it says 'or do we', which means it is an alternative way of looking at something. Hence, **EACBDF** is the right sequence.

The correct answer is Option (2)

Q32

Nourish

Solution

Dupe, Swindle and Hoax are synonyms of Gull.

Q33

ardent

Solution

Blase means unimpressed with or indifferent to something . Fiery means passionate.Intriguing means arousing one's curiosity. Ardent means very enthusiastic or passionate. So ardent is the answer.

Q34

Carefree

Solution

Blithe : happy or carefree

Emotive : arousing intense feelings.

Distress : Extreme anxiety - Antonym

Hence the answer is Carefree

Q35

would have

Solution

The sentence structure for the if-conditional clause is "if I had ..... I would have".

The appropriate verb would be "would have".

Q36 was watching

Solution

had watched and was watching are suitable in the blank. But had watched is not contextually correct. One could have done the homework while watching TV. Hence, 'was watching' is the correct answer.

Q37 Does he know

Solution

The word does is in simple present tense, and so thought would not be correct. Part c is equally confusing but the question mark helps us to eliminate the option. Having a sentence like " Will he have to face the consequences.The answer is "does he know"

Q38 who demanded that it

Solution

'Demanding' is incorrect as it sounds as if the action is taking place now rather than during WW1.

"and their demand to" changes the meaning and introduces an unclear pronoun, 'their'. "who demanded them to" also has an unclear pronoun", 'them'.

If the pronoun refers to the Russian state, it should be singular. Only "who demanded that it" is correct.

Q39 an

Solution

The word hour has a vowel sound. Thus takes the article **an**.

Q40 than

Solution

The correct phrase is "No sooner had I reached the office **than** I got the news of your arrival." The expression "no sooner... than" is used to indicate that one event happened immediately after another.

Q41. P

Solution

P/Football	R/Hockey	M/Golf	O/Badminton	S/Cricket	T/Chess	Q/Baseball	N/Tennis

P sits on the immediate left of R.

Q42. Second to the right

Solution

P/Football	R/Hockey	M/Golf	O/Badminton	S/Cricket	T/Chess	Q/Baseball	N/Tennis

N is second to the right of who likes Chess.

Q43. Two

Solution

P/Football	R/Hockey	M/Golf	O/Badminton	S/Cricket	T/Chess	Q/Baseball	N/Tennis

Two persons are there between M and T.

Q44. Badminton

Solution

P/Football	R/Hockey	M/Golf	O/Badminton	S/Cricket	T/Chess	Q/Baseball	N/Tennis

O likes Badminton game.

Q45. M

Solution

P/Football	R/Hockey	M/Golf	O/Badminton	S/Cricket	T/Chess	Q/Baseball	N/Tennis

M likes Golf.

Q46. 1

Solution

ended	2
or	0
your	6
fast	7
left	1
journey	5
the	4
behind	9
move	3
life	8

Q47. Move

Solution

ended	2
or	0
your	6
ast	7
eft	1
ourney	5
he	4
ehind	9
nove	3
fe	8

Q48. 5 6 8

Solution

ended	2
or	0
your	6
fast	7
left	1
journey	5
the	4
behind	9
move	3
life	8

Q49. 7

Solution

ended	2
or	0
your	6
fast	7
left	1
journey	5
the	4
behind	9
move	3
life	8

Q50. None is true

Solution

M is the brother of P. P is the daughter of Q and Q is the sister of R

ɹs, P is the niece of R

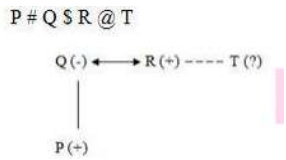
is the brother of P

s the mother of P

ɹs, None of the options are true.

Q51. P is the daughter of R

Solution



(-) = Male (+) = Female

Thus, P is daughter of R

Q52. relationship between parents is strained.

Solution

Refer to para 1, in the sentence - 'When ...insecure'. Since the child's insecurity is because of the parents' strained relationship, "relationship between parents is strained." is the appropriate choice.

Q53. reading or collecting things.

Solution

Refer to the fourth sentence of the last paragraph. "reading or collecting things."is correct.

Q54. the mental health of the gifted students.

Solution

The focus of the passage is clearly mentioned in the opening sentence of the passage. Hence "the mental health of the gifted students." is the right choice.

Q55. different from his classmates and cannot share his interests with them.

Solution

The first part of the last paragraph clearly explains why the gifted child needs help in human relations. Only "different from his classmates and cannot share his interests with them." presents this idea While options "predisposed to solitary pursuits" & "emotionally imbalanced" are irrelevant, option "inadequate in physical activities." is the effect of what happens if the child does not receive help.

So the right option is "different from his classmates and cannot share his interests with them".

Q56. has the opportunity to notice maladjustments in a child.

Solution

e author says that the teachers have the opportunity to observe the behavior of students. So, a cher's role becomes important in understanding the students (Para 1 sentence 2). Hence only :ion "has the opportunity to notice maladjustment in a child."is correct.

Q57. self-sufficient

Solution

The context of the passage is about production and the usefulness of Green Revolution. Hence the appropriate word would be self-sufficient

Q58. making technological

Solution

The word to look out in the passage is advancement and the word logically filling the blank would be making technological

Q59. truth is far from it

Solution

The hint of completing the sentence would be the conjunction "but" it refers to contradiction of the idea mentioned earlier. The opening of the passage talks about the benefits of green revolution. With the introduction of the conjunction, the answer option should be negative. The only possible option to fit in would be truth is far from it.

Q60. to be gearing up

Solution

According to the passage, The Indian Economy is suppose to be prepared to take on the world but we are still surrounded by the adversities of farmer suicide.
The appropriate choice would be "to be gearing up "

Section 2 - Pseudocode and Computer Fundamentals

Q1 All of the mentioned options

Solution

No Solution

Q2 128 bits

Solution

No Solution

Q3 Larger address space

Solution

No Solution

Q4 End-to-End Communication

Solution

No Solution

Q5 Synchronize sequence numbers

Solution

No Solution

Q6 It is faster than TCP

Solution

No Solution

Q7 Low Overhead

Solution

No Solution

Q8 Reliable Data Delivery

Solution

No Solution

Q9 UDP

Solution

No Solution

Q10 End-to-End Communication

Solution

No Solution

Q11 DNS resolver

Solution

No Solution

Q12 MAC Address Resolution

Solution

No Solution

Q13

Application Layer

Solution

No Solution

Q14

53

Solution

No Solution

Q15

34

Solution

In funn(12, 6) , if condition true so funn(12-10,12+10) is called i.e funn(2,22)

In funn(2, 22) , if condition gets false so it returns 22+2+10 => 34.

Q16

6

Solution

In funn(5,2) => if block gets executed and call funn(1,4)

In funn(1,4) => function return 1+4+1 => 6

Q17

39

Solution

q=7+5+7=19

r=4+5+7=16

Then if block condition gets failed, so 4+19+16=39

Q18

12

Solution

b=(7^12)&2 => 2

b=9&2 => 0

If block condition gets failed, so 2+0+10 = 12.

Q19

None of the mentioned options

Solution

Always a+10> b so function called infinite time recursively.

Q20

4

Solution

Initially p=0, q=4, r=8,

In for loop when r=3, if block gets true and breaks the loop. So 0+4=>4 gets printed.

Q21

8

Solution

In funn(1,4)=> funn(2,3)+funn(4,5)

in funn(2,3)=> return 3.

in funn(4,5)=> return 5.So in funn(1,4) 3+5 = 8 gets printed.

Q22

29

Solution

p=12&8+1=9

q=(9^6)&9= 9

r=2+9=11

Therefore 9+9+11=29

Q23

24

Solution

In function(4): 4*function(3)

In function(3): 3*function(2)

In function(2): 2*function(1)

In function(1): 1*function(0)

In function(0): return 1 to function(1)

So function(1) = 1*1 = 1 return to function(2)

So function(2) = 2*1 = 2 return to function(3)

So function(3) = 3*2 = 6 return to function(4)

So function(4) = 4*6 = 24.Prints 24.

Q24

25

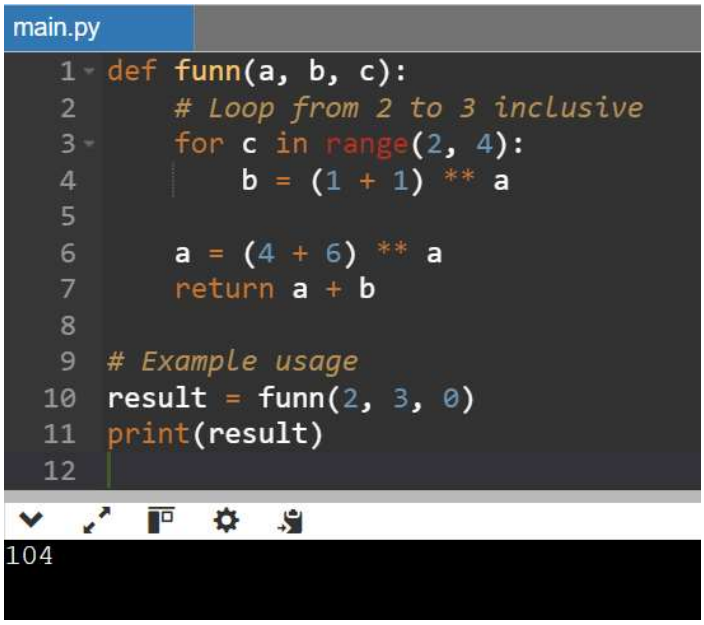
Solution

Here the value of pp updated to 15,qq updated to 4 and rr=6 after performing the bitwise operations. So output is 15+4+6 => 25.

Q25

104

Solution



Q26

It will print three If x=9, y=8, z=6

Solution

No Solution

Q27

39

Solution

No Solution

Q28

3

Solution

Initially q=81,r=10.

Iteration1: 81 > 10 (true) so p=91,s=9,t=81,q=27

Iteration2: 27 > 10 (true) so p=37,s=3,t=9,q=9

Iteration3: 9 > 10 (false) So again checks 9>3 (true so 3 gets printed)

Q29

23

Solution

Initially a=8, b=2, c=4, if block condition gets failed so b value updated to 11. So 8+11+4=> 23

Q30

9

Solution

No Solution

Section 3 - Coding

Q1

Test Case

Input

```
1 2 5 10
150
```

Output

```
15
```

Weightage - 10

Input

```
2 4 6
190
```

Output

```
32
```

Weightage - 15

Input

```
3 5 9 11
25
```

Output

```
3
```

Weightage - 25

Input

```
1 4 6 8
15
```

Output

```
3
```

Weightage - 15

Input

```
1 3 4 7 9 13 15 17 19 25
509
```

Output

```
21
```

Weightage - 25

Input

```
5 3 2 1
50
```

Output

```
10
```

Weightage - 10

Sample Input

```
1 2 5
11
```

Sample Output

```
3
```

Sample Input

```
1
0
```

Sample Output

```
0
```

Sample Input

```
2
3
```

Sample Output

```
-1
```

Solution

```java
import java.util.Arrays;
import java.util.Scanner;

public class Main {
    public static int coinChange(int[] coins, int amount) {
        Arrays.sort(coins);  // Sort the coin denominations in ascending order
        int count = 0;

        for (int i = coins.length - 1; i >= 0; i--) {
            while (amount >= coins[i]) {
                amount -= coins[i];
                count++;
            }
        }

        return amount == 0 ? count : -1;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        //System.out.print("Enter the denominations of coins (space-separated): ");
        String coinsInput = scanner.nextLine();
        String[] coinsStr = coinsInput.split(" ");
        int[] coins = new int[coinsStr.length];
        for (int i = 0; i < coinsStr.length; i++) {
            coins[i] = Integer.parseInt(coinsStr[i]);
        }

        //System.out.print("Enter the amount: ");
        int amount = scanner.nextInt();

        int result = coinChange(coins, amount);
       System.out.println(result);
    }
}
```

```cpp
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int coinChange(vector<int>& coins, int amount) {
    sort(coins.begin(), coins.end()); // Sort the coin denominations in ascending order
    vector<int> dp(amount + 1, amount + 1); // Initialize the dp array
    dp[0] = 0; // Base case: zero amount requires zero coins

    for (int i = 1; i <= amount; ++i) {
        for (int j = 0; j < coins.size(); ++j) {
            if (coins[j] <= i) {
                dp[i] = min(dp[i], dp[i - coins[j]] + 1);
            } else {
                break; // Since coins are sorted, no need to continue if current coin is greater than amount
            }
        }
    }

    return dp[amount] > amount ? -1 : dp[amount]; // If dp[amount] is still greater than amount, it means amount cannot be formed
}

int main() {
    vector<int> coins;
    int coin;
    char c;
```

```
    // Read the denominations of coins
    while (cin >> coin) {
        coins.push_back(coin);
        if (cin.peek() == '\n') break;
    }

    int amount;
    cin >> amount; // Read the amount

    int result = coinChange(coins, amount);
    cout << result << endl; // Print the result

    return 0;
}
```

Q2

Test Case

Input	Output
algorithm algorhythm	2

Weightage - 15

Input	Output
exponential polynomial	6

Weightage - 25

Input	Output
initialization optimization	5

Weightage - 25

Input	Output
cat cut	1

Weightage - 10

Input	Output
sunday saturday	3

Weightage - 10

Input	Output
great greek	2

Weightage - 15

Sample Input

```
horse
ros
```

Sample Output

```
3
```

Sample Input

```
intention
execution
```

Sample Output

```
5
```

Solution

```java
import java.util.Scanner;

class Main {
    public static int minDistance(String word1, String word2) {
        int m = word1.length();
        int n = word2.length();

        // Create a 2D array to store the minimum edit distance
        int[][] dp = new int[m + 1][n + 1];

        // Initialize the first row and column
        for (int i = 0; i <= m; i++) {
            dp[i][0] = i;
        }
        for (int j = 0; j <= n; j++) {
            dp[0][j] = j;
        }

        // Calculate the minimum edit distance using dynamic programming
        for (int i = 1; i <= m; i++) {
            for (int j = 1; j <= n; j++) {
                if (word1.charAt(i - 1) == word2.charAt(j - 1)) {
                    // Characters are equal, no operation needed
                    dp[i][j] = dp[i - 1][j - 1];
                } else {
                    // Choose the minimum of insert, delete, or replace operation
                    dp[i][j] = 1 + Math.min(dp[i - 1][j - 1], Math.min(dp[i][j - 1], dp[i - 1][j]));
                }
            }
        }

        return dp[m][n];
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        //System.out.print("Enter word1: ");
        String word1 = scanner.nextLine();

        //System.out.print("Enter word2: ");
        String word2 = scanner.nextLine();

        int minDistance = minDistance(word1, word2);
        System.out.println(minDistance);

        scanner.close();
    }
}
```

```cpp
#include <iostream>
#include <string>
#include <algorithm>
```

```
using namespace std;

int minDistance(const string& word1, const string& word2) {
    int m = word1.length();
    int n = word2.length();

    // Create a 2D array to store the minimum edit distance
    int dp[501][501] = {0};

    // Initialize the first row and column
    for (int i = 0; i <= m; i++) {
        dp[i][0] = i;
    }
    for (int j = 0; j <= n; j++) {
        dp[0][j] = j;
    }

    // Calculate the minimum edit distance using dynamic programming
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (word1[i - 1] == word2[j - 1]) {
                // Characters are equal, no operation needed
                dp[i][j] = dp[i - 1][j - 1];
            } else {
                // Choose the minimum of insert, delete, or replace operation
                dp[i][j] = 1 + min(dp[i - 1][j - 1], min(dp[i][j - 1], dp[i - 1][j]));
            }
        }
    }

    return dp[m][n];
}

int main() {
    // Get input words
    string word1, word2;
    getline(cin, word1);
    getline(cin, word2);

    int minDistanceValue = minDistance(word1, word2);
    cout<< minDistanceValue << endl;

    return 0;
}
```